How Do Keyboard Type, Background Music, and Text Format Affect Typing Speed?

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Question

The motivation for this experiment comes from how widely typing is used in everyday school and work activities. Since typing speed can directly affect productivity, our project investigates how keyboard type, background music, and text format influence typing performance. By studying these factors and their interactions, we hope to identify which combinations lead to better typing performance.

Plan

Our main goal is to determine whether participants type faster using a mechanical keyboard or an Apple keyboard, whether background music affects typing speed, and whether typing a complete paragraph or a list of unrelated words changes performance. We are also interested in possible interaction effects. For example, background music may affect typing speed differently depending on which keyboard is being used.

This project is meaningful because understanding how these factors influence typing efficiency may help people create a more productive working or studying environment. To ensure that the results reflect actual typing performance rather than participants simply focusing on speed, all typed text was required to be error-free.

Our response variable is typing speed, measured in words per minute, or WPM. The experimental unit is each individual typing trial. This is completed by a participant under a specific combination of keyboard type, background music, and text format. Each trial produces one WPM measurement.

The experiment includes three design factors, with two levels for each factor:

- Keyboard type (Factor A): Mechanical keyboard (-1) or Apple keyboard (1)
- Background music (Factor B): Silence (-1) or music (1)

- Text format (Factor C): Random words (-1) or paragraph (1)

The blocking factor is the participant. Since each participant has a different typing ability, blocking helps control for these individual differences. This allows us to focus more clearly on the effects of the three experimental factors.

For the experimental setup, we used a (2^3) full factorial design. This means that three factors were tested across eight total treatment combinations. Each treatment combination was repeated three times by each of the three participants. Therefore, each participant completed 24 runs, resulting in (8 *3 * 3 = 72) total trials.

We also randomized the run order for participants and alternated between different treatment combinations to reduce the effects of learning, fatigue, and the order in which the trials were completed.

Data

Data Collection Process

Data were collected from three participants, and each participant therefore completed 24 typing trials as described above. For each trial, the participant followed the treatment order generated through randomization. The assigned keyboard was first connected and prepared, and the background condition was set to either silence or with music. The participant then completed either a random-word test or a paragraph test using Monkeytype. Each trial lasted 30 seconds, and the participant's typing speed in words per minute was recorded after the trial. The website, test duration, and general procedure were kept consistent across all observations.

For the music condition, participants listened to *River Flows in You* by Yiruma using AirPods. We chose a steady instrumental track with no lyrics because it provided consistent background music without being too distracting. The same song was used throughout the

experiment, and the volume was maintained at the same level. In the silent condition, no music was played during the trial.

The two text-format conditions were designed to represent different typing tasks. In the random-word condition, participants typed a sequence of unrelated words, such as “table river window market glass”. In the paragraph condition, participants typed connected sentences with a meaningful structure. An example was “typing is commonly used in school and office settings”. The texts were kept as similar as possible in both length and typing difficulty. All 24 paragraphs contained exactly 100 words and used a similar level of vocabulary and sentence complexity. For the random-word condition, each set also contained 100 words. We used the same words in every set but arranged them in a different random order. This helped keep the word length and overall difficulty consistent across all 72 runs. Capitalization and punctuation were removed to reduce variation caused by the use of the Shift key and punctuation keys. We also attempted to use familiar words with similar average word lengths across the passages. Participants did not get the chance to look at the prepared random words or paragraph before their tests. A different text was used for each trial to reduce memorization and prevent participants from improving simply because they had previously seen the same passage.

Participants were required to maintain 100% typing accuracy, meaning that participants needed to correct errors during the test. There were no missing observations in the final dataset. Please see our collected data in *Appendix A*.

Initial Analysis of Data

An initial summary of the observations (see Figure 1 below) showed substantial differences in baseline typing speed among participants. The median WPM varied by approximately 20 WPM across the three participants, with Participant 3 generally typing more

slowly than Participants 1 and 2. This large between-participant difference supported the decision to use participants as a blocking factor.

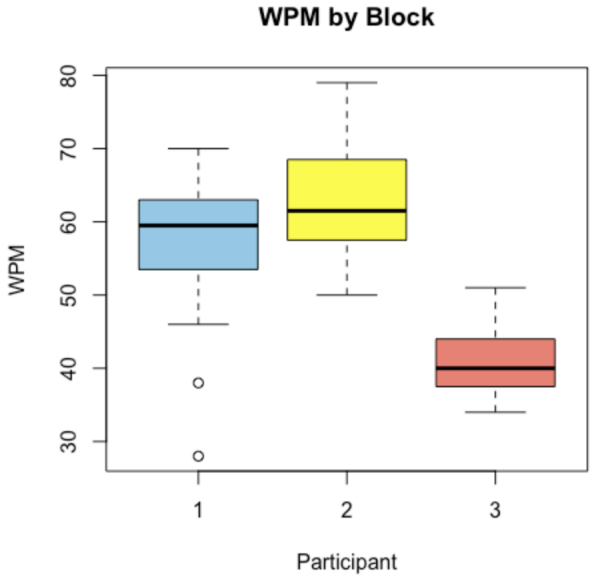


Figure 1. WPM by Participant

When the data were grouped by treatment factor, the largest visible difference was between the two keyboard types, with a difference of approximately 7 WPM in median typing speed. In comparison, the WPM distributions under music and silence were highly similar, as were the distributions for random words and paragraphs.

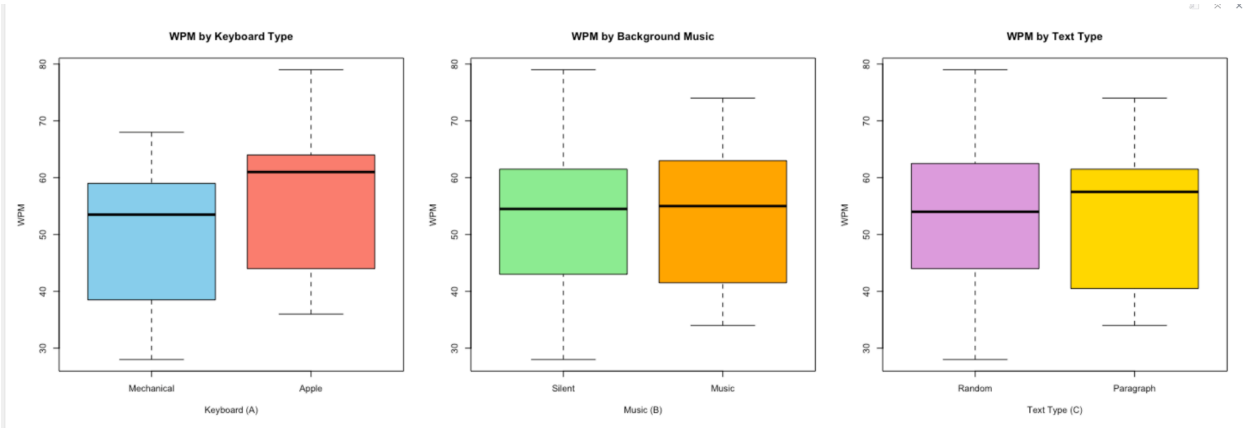


Figure 2. Boxplot of Average WPM by Factor

Analysis

Statistical Model and Model Selection

We first fitted a full model containing the main effects of participant, keyboard type, background sound, and text format, as well as all possible interactions among these factors. The interactions involving participants (i.e. blocks) were included to examine whether the effects of keyboard, sound, and text format differed across participants. Please see the full effects model equation in *Appendix B*. The full Model yields the ANOVA Table below:

```
model_full <- aov(y ~ Block * A * B * C, data = data)
summary(model_full)
```

```
             Df Sum Sq Mean Sq F value   Pr(>F)
Block         2   6293  3146.5  90.187  < 2e-16 ***
A             1   1005  1005.0  28.806 2.28e-06 ***
B             1      3     3.1   0.090   0.7660
C             1      0     0.3   0.010   0.9209
Block:A       2    155    77.3   2.217   0.1200
Block:B       2     60    30.0   0.861   0.4291
A:B           1     74    74.0   2.121   0.1518
Block:C       2    248   123.8   3.547   0.0366 *
A:C           1      6     6.1   0.176   0.6771
B:C           1      3     3.1   0.090   0.7660
Block:A:B     2     79    39.3   1.128   0.3322
Block:A:C     2     23    11.3   0.324   0.7251
Block:B:C     2    180    90.0   2.581   0.0862 .
A:B:C         1     45    45.1   1.293   0.2611
Block:A:B:C   2    169    84.3   2.416   0.1001
Residuals    48   1675    34.9
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1
```

Figure 3. Full Model ANOVA Table

We also fitted a reduced model that retained participants as a blocking factor but removed all interactions involving participants. The reduced model still included the main effects of keyboard type, background sound, and text format, together with all two-way and three-way interactions among the three treatment factors. Please see the reduced effects model equation in *Appendix C*. The reduced model yields the ANOVA Table below:

```
model_reduced <- aov(y ~ Block + A * B * C, data = data)
summary(model_reduced)
```

```
            Df Sum Sq Mean Sq F value   Pr(>F)
Block        2   6293  3146.5  75.412  < 2e-16 ***
A            1   1005  1005.0  24.087 7.01e-06 ***
B            1      3     3.1   0.075    0.785
C            1      0     0.3   0.008    0.928
A:B          1     74    74.0   1.774    0.188
A:C          1      6     6.1   0.147    0.703
B:C          1      3     3.1   0.075    0.785
A:B:C        1     45    45.1   1.081    0.302
Residuals   62   2587    41.7
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1
```

Figure 4. Reduced Model ANOVA Table

As shown in the model comparison output (see Figure 5 below), the full model did not significantly improve fit over the reduced model, $F(14,48)=1.868$, $p=0.0552$. Therefore, at the 5% significance level, the block interaction terms were pooled into the error term and the reduced model was used for the remaining analysis.

```
anova(model_reduced, model_full)
```

```
Analysis of Variance Table

Model 1: y ~ Block + A * B * C
Model 2: y ~ Block * A * B * C
  Res.Df    RSS Df Sum of Sq      F  Pr(>F)
1     62 2586.9
2     48 1674.7 14    912.25 1.8677 0.05521 .
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1
```

Figure 5. Model Comparison

Main Effects

The ANOVA table for the reduced model (see Figure 4 above) shows that keyboard type is the only factor with a statistically significant main effect on typing speed, $F(1,62) = 24.087$, $p < 0.001$. This is consistent with the roughly 7 WPM difference in median typing speed observed

between the two keyboards in Figure 2. Background sound ($F(1,62) = 0.075$, $p = 0.785$) and text format ($F(1,62) = 0.008$, $p = 0.928$) show no evidence of an effect on typing speed.

Interaction Effects

None of the two-way or three-way interactions among keyboard type, background sound, and text format were statistically significant (all $p > 0.18$; see the reduced model ANOVA table above). Figure 6 below shows the estimated mean WPM for each pair of factors. In all three panels, the lines are roughly parallel, meaning that the effect of keyboard type does not meaningfully change depending on whether music is playing or which text format is used, and vice versa.

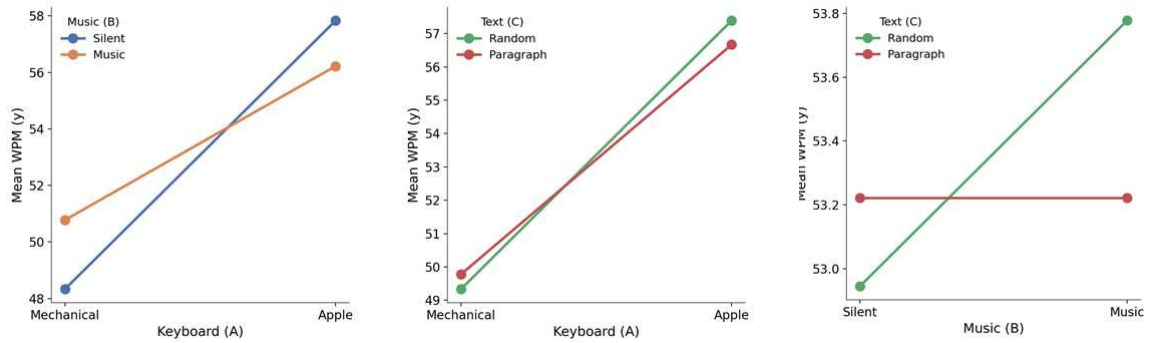


Figure 6. Two-way interaction plots: Keyboard × Sound, Keyboard × Text, and Sound × Text

Model Diagnostics

Before finalizing our conclusions, we checked the assumptions underlying the ANOVA using the residuals from the reduced model, shown in Figure 7 below.

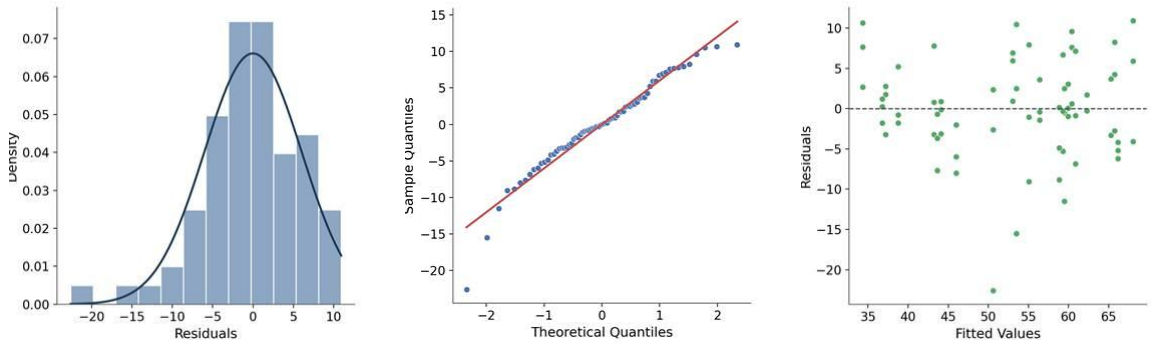


Figure 7. Residual histogram, normal Q-Q plot, and residuals vs. fitted values for the reduced model

The residuals are randomly spread around zero with no specific patterns, indicating linearity and constant variance. Additionally, the QQ-plot mostly follows a straight line with some volatility toward the tails, representing a rough normality which is also supported by the histogram. This suggests some caution when interpreting borderline p-values, but the conclusion regarding keyboard type still holds.

Run Order Analysis

The run order was randomized separately for each participant to distribute possible practice and fatigue effects across the treatment conditions. To assess whether typing speed changed systematically over time, WPM was plotted against run order for each participant (See Figure 8 below). Although there was noticeable trial-to-trial variation, none of the three participants showed a clear and consistent upward or downward trend across the 24 trials. Therefore, the graph provides no strong visual evidence that practice or fatigue substantially affected typing speed during the experiment.

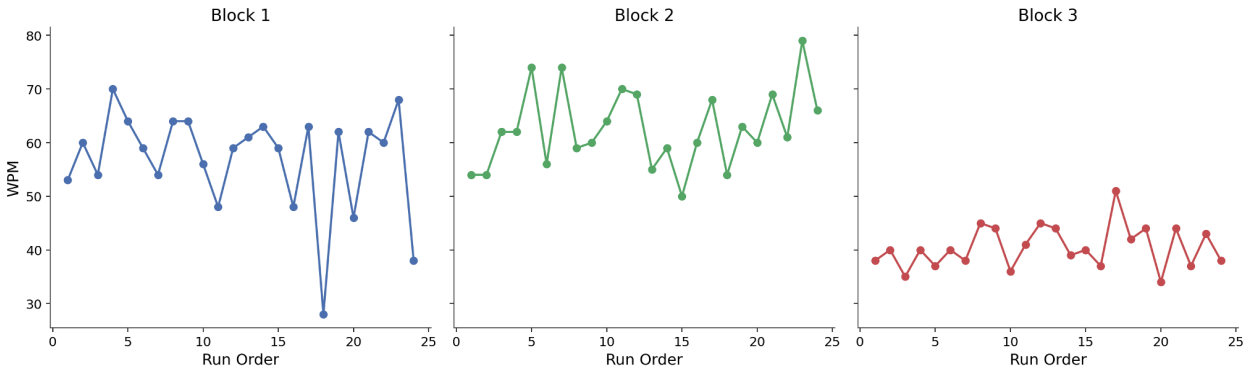


Figure 6. Data vs Run Order By Blocks

Conclusion

Overall, our analysis shows that keyboard type is the primary driver of typing speed in this experiment: participants typed about 7 WPM faster on the Apple keyboard than on the Mechanical keyboard, and this advantage held regardless of whether background music was playing or which text format participants typed. Background music and text format did not have a measurable effect on typing performance, either on their own or in combination with keyboard type.

Limitations

Limitation of this experiment is the small sample size, as only three participants were included. Although each participant completed multiple trials, the results may not represent people with different typing abilities, keyboard familiarity, or experience levels. The findings should therefore be interpreted mainly in the context of the participants in this study.

Another limitation is that each typing test lasted only 30 seconds. This may not accurately reflect typing performance during longer tasks, such as writing assignments or emails, where sustained speed, concentration, and fatigue may become more important.

The music condition was also limited to one instrumental piece, River Flows in You by Yiruma. Since the song has no lyrics and a relatively calm style, the results may not apply to

music with lyrics, faster tempos, or other genres. Participants' familiarity with or preference for the song may also have influenced their concentration and typing performance.

Future Improvement

Future studies could include more participants with different typing abilities and levels of keyboard experience. This would improve the generalizability of the results and allow researchers to examine whether familiarity with a keyboard affects typing performance.

Longer typing tests, such as several minutes rather than 30 seconds, could also better reflect real-world typing tasks and provide a more reliable measure of sustained speed and accuracy.

Future experiments could also compare several music conditions, such as instrumental music, music with lyrics, and different genres or tempos, while keeping the volume consistent. Participants' familiarity with and preference for the music could also be recorded to determine whether these factors influence performance.

Appendix

Appendix A - Collected Data

Blocks (Participant)	Keyboard Type	Background Music	Text Type	WPM (Response)
1	-1	-1	-1	53
1	1	1	1	60
1	-1	1	-1	54
1	1	-1	1	70
1	-1	1	1	64
1	1	1	-1	59
1	-1	-1	1	54
1	1	-1	-1	64
1	1	-1	-1	64
1	-1	1	1	56
1	1	1	-1	48
1	-1	-1	1	59
1	1	-1	1	61
1	-1	1	-1	63
1	1	1	1	59
1	-1	-1	-1	48
1	1	1	1	63
1	-1	-1	-1	28
1	1	-1	-1	62
1	-1	1	-1	46
1	1	1	-1	62
1	-1	-1	1	60
1	1	-1	1	68
1	-1	1	1	38
2	-1	1	1	54
2	-1	1	-1	54
2	1	-1	1	62
2	1	1	-1	62
2	1	-1	-1	74
2	-1	-1	-1	56
2	1	1	1	74
2	-1	-1	1	59
2	-1	1	-1	60
2	1	-1	-1	64
2	1	1	1	70
2	1	1	-1	69
2	-1	-1	-1	55
2	-1	1	1	59
2	-1	-1	1	50
2	1	-1	1	60
2	-1	1	-1	68
2	-1	-1	1	54
2	1	1	1	63
2	-1	-1	-1	60
2	1	1	-1	69
2	1	-1	1	61
2	1	-1	-1	79
2	-1	1	1	66
3	1	-1	-1	38
3	-1	1	1	40
3	-1	-1	1	35
3	1	1	-1	40
3	-1	-1	-1	37
3	1	1	1	40
3	-1	1	-1	38
3	1	-1	1	45
3	-1	1	-1	44
3	1	1	1	36
3	1	-1	1	41
3	-1	-1	-1	45
3	1	1	-1	44
3	-1	1	1	39
3	1	-1	-1	40
3	-1	-1	1	37
3	1	1	-1	51
3	-1	-1	-1	42
3	1	-1	1	44
3	-1	1	1	34
3	1	-1	-1	44
3	-1	1	-1	37
3	1	1	1	43
3	-1	-1	1	38

Appendix B - Full Effects Model Equation

$$
\begin{aligned}
Y_{ijklr} = \mu &+ \rho_i + \alpha_j + \beta_k + \gamma_l \\
&+ (\rho\alpha)_{ij} + (\rho\beta)_{ik} + (\rho\gamma)_{il} + (\alpha\beta)_{jk} + (\alpha\gamma)_{jl} + (\beta\gamma)_{kl} \\
&+ (\rho\alpha\beta)_{ijk} + (\rho\alpha\gamma)_{ijl} + (\rho\beta\gamma)_{ikl} + (\alpha\beta\gamma)_{jkl} \\
&+ (\rho\alpha\beta\gamma)_{ijkl} + \varepsilon_{ijklr}
\end{aligned}
$$

Where:

Y_{ijklr} = observed typing speed, measured in words per minute, for participant i, keyboard level j, background-sound level k, text-format level l, and replication r.

μ = overall mean typing speed across all participants and experimental conditions.

ρ_i = effect of participant i.

α_j = main effect of keyboard type j.

β_k = main effect of background-sound condition k.

γ_l = main effect of text format l.

$(\rho\alpha)_{ij}$ = interaction between participant and keyboard type.

$(\rho\beta)_{ik}$ = interaction between participant and background sound.

$(\rho\gamma)_{il}$ = interaction between participant and text format.

$(\alpha\beta)_{jk}$ = interaction between keyboard type and background sound.

$(\alpha\gamma)_{jl}$ = interaction between keyboard type and text format.

$(\beta\gamma)_{kl}$ = interaction between background sound and text format.

$(\rho\alpha\beta)_{ijk}$ = interaction among participant, keyboard type, and background sound.

$(\rho\alpha\gamma)_{ijl}$ = interaction among participant, keyboard type, and text format.

$(\rho\beta\gamma)_{ikl}$ = interaction among participant, background sound, and text format.

$(\alpha\beta\gamma)_{jkl}$ = three-way interaction among keyboard type, background sound, and text format.

$(\rho\alpha\beta\gamma)_{ijkl}$ = four-way interaction among participant, keyboard type, background sound, and text format.

ε_{ijklr} = random experimental error.

Appendix C - Reduced Effects Model Equation

$$
\begin{aligned}
Y_{ijklr} = \mu + \rho_i + \alpha_j + \beta_k + \gamma_l \\
+ (\alpha\beta)_{jk} + (\alpha\gamma)_{jl} + (\beta\gamma)_{kl} \\
+ (\alpha\beta\gamma)_{jkl} + \varepsilon_{ijklr}
\end{aligned}
$$

Where:

Y_{ijklr} = observed typing speed, measured in words per minute, for participant i, keyboard level j, background-sound level k, text-format level l, and replication r.

μ = overall mean typing speed across all participants and experimental conditions.

ρ_i = effect of participant i.

α_j = main effect of keyboard type j.

β_k = main effect of background-sound condition k.

γ_l = main effect of text format l.

$(\alpha\beta)_{jk}$ = interaction between keyboard type and background sound.

$(\alpha\gamma)_{jl}$ = interaction between keyboard type and text format.

$(\beta\gamma)_{kl}$ = interaction between background sound and text format.

$(\alpha\beta\gamma)_{jkl}$ = three-way interaction among keyboard type, background sound, and text format.

ε_{ijklr} = random experimental error.